

WEEK – 3

Java programming -Java Buzzwords, Data types, Variables, Operators, Control structures, Arrays, Console input and output

Sample Program:

Write a java program which reads marks of five subjects through command line and print the total and average.

Steps

1. Start the program.
2. Import the Scanner class from java.util.
3. Create a Scanner object to accept input from the user.
4. Initialize total to 0.
5. Display a message asking the user to enter 5 marks.
6. Use a for loop to read the marks of five subjects.
7. Add each mark to total.
8. Calculate the average using:

$\text{average} = \text{total} / 5.0$

9. Display the total marks.
10. Display the average marks.
11. Close the Scanner object.
12. Stop the program.

Procedure

1. Create a Java class named Marks.
2. Define the main() method.
3. Create a Scanner object using:

```
Scanner sc = new Scanner(System.in);
```

4. Declare and initialize:

```
int total = 0;
```

5. Use a for loop from 1 to 5.
6. Read each subject's marks using `sc.nextInt()`.

7. Add the entered marks to total.

8. Calculate the average:

`double average = total / 5.0;`

9. Print the total and average.

10. Close the Scanner and terminate the program.

Algorithm

Step 1: Start.

Step 2: Create a Scanner object.

Step 3: Initialize total = 0.

Step 4: Repeat for i = 1 to 5.

Step 5: Read the marks.

Step 6: Add the marks to the total:

`total = total + marks`

Step 7: Calculate the average:

`average = total / 5.0`

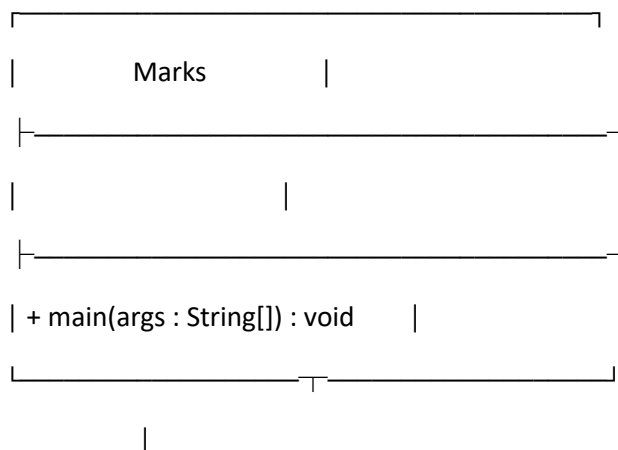
Step 8: Display the total marks.

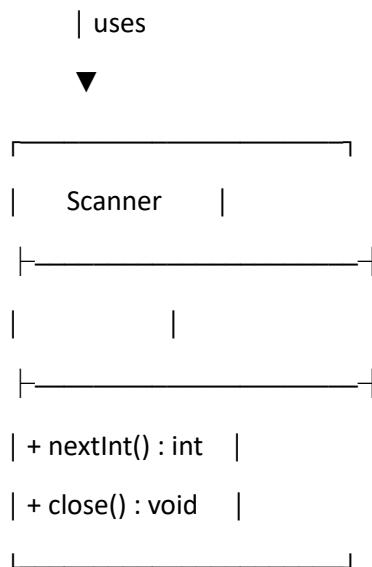
Step 9: Display the average marks.

Step 10: Close the Scanner.

Step 11: Stop.

UML Class Diagram





UML Explanation

Element	Description
Marks	User-defined class
main()	Main method of the program
String[] args	Command-line argument
void	Return type of main()
Scanner	Java class used to read input
nextInt()	Reads integer marks
close()	Closes Scanner

Code:

```

package core_java;

import java.util.Scanner;

public class Marks {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int total = 0;
  
```

```
System.out.println("Enter 5 marks:");
```

```
for (int i = 1; i <= 5; i++) {  
    int marks = sc.nextInt();  
    total = total + marks;  
}
```

```
double average = total / 5.0;
```

```
System.out.println("Total Marks = " + total);
```

```
System.out.println("Average Marks = " + average);
```

```
sc.close();
```

```
}
```

```
}
```

Output:



The screenshot shows a Java console window titled "Console" with a close button. The window displays the following text:

```
<terminated> Marks [Java Application] C:\Program Files\Java\jdk-25.0.3\bin\javaw.exe (13 Aug 2026, 10:04:40 pm - 10:05:00 pm elapsed 0:00:20) [pid: 16356]  
Enter 5 marks:  
88  
78  
98  
85  
65  
Total Marks = 414  
Average Marks = 82.8
```

The input marks (88, 78, 98, 85, 65) are displayed in green text, while the output (Total Marks = 414, Average Marks = 82.8) is in black text.